

Binding of the Priming Nucleotide in the Initiation of Transcription by T7 RNA Polymerase*

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Unlike DNA polymerases, an RNA polymerase must initiate transcription *de novo*, that is binding of the initiating (+1) nucleoside triphosphate must be achieved without benefit of the cooperative binding energetics of an associated primer. Since a single Watson-Crick base pair is not stable in solution, RNA polymerases might be expected to provide additional stabilizing interactions to facilitate binding and positioning of the initiating (priming) nucleoside triphosphate at position +1. Consistent with base-specific stabilizing interactions, of the 17 T7 RNA polymerase promoters in the phage genome, 15 begin with guanine. In this work, we demonstrate that the purine N-7 is important in the utilization of the initial substrate GTP. The fact that on a template encoding AG as the first two bases in the transcript (as in the remaining two of the T7 genome) transcription starts predominantly (but not exclusively) at the G at position +2 additionally implicates the purine O-6 as an important recognition element in the major groove. Finally, results suggest that these interactions serve primarily to position the initiating base in the active site. It is proposed that T7 RNA polymerase interacts directly with the Hoogsteen side of the initial priming GTP (most likely via an interaction with an arginine side chain in the protein) to provide the extra stability required at this unique step in transcription.

The initiation of transcription imposes some unique mechanistic requirements on an RNA polymerase. In contrast to events occurring during elongation, at the initial step of transcription initiation, two substrate nucleoside triphosphate molecules must position accurately in the active site. Clearly a part of the binding energetics is derived from Watson-Crick interactions between the incoming bases and those in the template strand of DNA, but just as clearly, base pairing interactions are not sufficient to provide the binding energetics required for full function. Indeed, a single Watson-Crick base pair is unstable in solution (1).

It is understood that for the Watson-Crick placement of the elongating nucleotide (position +2 at initiation), additional energy for binding of the nucleotide comes from interactions between its triphosphate, magnesium, and protein functional

groups (2–5). This interaction would not be expected to be important in binding the initiating (+1) nucleotide, and it has been shown in the T7 system that guanosine monophosphate and even the nucleoside guanosine have K_m values comparable to or lower than that of the triphosphate GTP (6). Some additional interaction(s) must be at play.

Most RNA polymerases show some preference for the initial base of the transcript. *Escherichia coli* RNA polymerase promoters often initiate with ATP, although some promoters begin with other NTPs at the first position in the transcript (7, 8). Of the 17 phage RNA polymerase promoters in the T7 genome, the canonical +1 position of 15 begins with GTP, while two promoters begin with ATP (9). Recent studies have demonstrated that T7 RNA polymerase initiates poorly on promoters encoding A at position +1; transcription instead initiates predominantly with an encoded G at position +2 (10–12). The SP6 enzyme similarly demonstrates a 5–20-fold reduction in the level of RNA production on promoters lacking an encoded G at position +1 (13).

The preference for initiation with GTP might suggest the importance of a strong Watson-Crick base pairing interaction between the substrate and the templating base (GC pairs being generally stronger than AT pairs), but the lack of promoters that initiate with CTP might suggest that this is not an important criterion. A remaining possibility is an interaction of the protein with the incoming base itself. This would explain the preference of a particular RNA polymerase for initiating with specific bases. Indeed, a recent study has implicated His-784 in contacting the 2-amino group of guanine in the minor groove (12).

In the current work, we demonstrate that the preference for GTP as the initiating nucleoside triphosphate by T7 RNA polymerase is indeed likely the result of base-specific, non-Watson-Crick interactions. We demonstrate that the N-7 and possibly O-6 positions along the major groove of guanine are specifically involved in positioning the substrate at position +1, explaining further the preference of this enzyme to initiate with GTP and consistent with a preference for a purine over a pyrimidine.

MATERIALS AND METHODS

RNA Polymerase—T7 RNA polymerase was prepared from *E. coli* strain BL21 carrying the overproducing plasmid pAR1219 (kindly supplied by F. W. Studier), which contains the T7 RNA polymerase gene under inducible control of the *lacUV5* promoter. The enzyme was purified, and its concentration was determined ($\epsilon_{280} = 1.4 \times 10^5 \text{ M}^{-1} \text{ cm}^{-1}$) as described previously (14). Purity of the enzyme was verified by SDS-PAGE.

Oligonucleotides—Oligonucleotides were synthesized by the phosphoramidite method on an Applied Biosystems Expedite 8909 DNA synthesizer. Single strands from a 1- μmol scale synthesis were purified trityl-on using an Amberchrom CG-161cd reverse phase resin (Toso-Haas Inc.) as described previously (15). Purity of the oligonucleotides was confirmed by denaturing (urea) gel electrophoresis of 5'-end-

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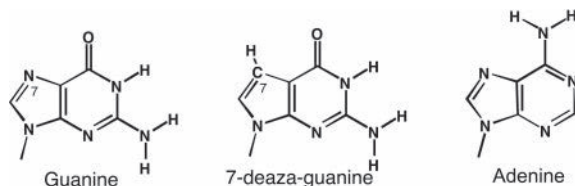


FIG. 1. Comparative structures of guanine, 7-deaza-guanine, and adenine.

labeled single strands. Double-stranded DNA was made by heating complementary single strands to 90 °C, allowing the resulting mixture to cool to room temperature over 2 h.

Kinetic Assays—Steady-state assays of transcription were carried out at 37 °C in a total volume of 20 μ l. The resulting mixture contained 30 mM HEPES (pH 7.8), 15 mM magnesium acetate, 25 mM potassium glutamate, 0.25 mM EDTA, 0.05% (v/v) Tween 20 (Calbiochem, protein grade), 0.8 mM GTP or 7-deaza-GTP, 0.4 mM ATP, CTP, and UTP each, and less than 0.06 μ M [α -³²P]GTP, [α -³²P]ATP, or [γ -³²P]ATP (PerkinElmer Life Sciences) as a label. Reactions also contained 0.2 μ M promoter DNA, 0.2 μ M T7 RNA polymerase, and 0.4 mM GMP where indicated. Reactions were incubated at 37 °C for 10 min and stopped by addition of an equal volume of 95% formamide, 20 mM EDTA (pH 7.8) gel loading buffer. The 3.0- μ l aliquots were loaded onto a 7 M urea, 18% polyacrylamide sequencing gel. After electrophoresis for 2.5 h at 2000 V/50 watts, gels were dried and quantified using an Amersham Biosciences Storm 840 PhosphorImager. The percent fall-off was calculated for each band by taking the ratio of the intensity (I_i) of the band i corrected for the number of radioactive labels incorporated (I_i/n_i) and dividing by the sum of corrected intensities of all the bands length i and longer and multiplied by 100% (Equation 1).

$$(\text{Percent Fall-off})_i = \frac{(I_i/n_i)}{\sum_i (I_i/n_i)} \times 100\% \quad (\text{Eq. 1})$$

Product Assignment—To properly assign all of the products, at least three parallel reactions were performed for each promoter construct. Conditions were as above, and reactions contained either [α -³²P]GTP, [α -³²P]ATP, or [γ -³²P]ATP (PerkinElmer Life Sciences) for labeling purposes. For each band, the (molar) ratios of [α -³²P]GTP to [α -³²P]ATP and of [α -³²P]ATP to [γ -³²P]ATP incorporation were calculated. The bands were assigned on the basis of these ratios and the approximate length of the RNA products. Results are shown in Table I.

RESULTS

It has long been known that T7 RNA polymerase prefers to initiate with guanosine as the first base in the RNA transcript (16–18). Recent results have demonstrated a very strong preference for initiation with GTP (C in the template strand, resulting in a 5' G) with a minor ability to initiate with ATP (10, 12, 19).

It is well established that RNA polymerases, in general, face a unique challenge at initiation, positioning the priming (+1) nucleotide without the aid of stabilizing covalent and noncovalent interactions with an upstream heteroduplex. Does the polymerase specifically bind or position GTP as the incoming substrate in a manner unique to initiation?

The Guanine N-7 of the Initiating Nucleotide Is Critical—The most likely base-specific contacts with G within a GC pair lie along the Hoogsteen face (20). To test the importance of a potential Hoogsteen contact in the RNA polymerase substrate GTP, we have carried out transcription in the presence of 7-deaza-GTP.

As illustrated in Fig. 1, this analog is very similar to GTP with the exception that the nitrogen at the 7 position is replaced by a carbon (and the nitrogen lone pair is replaced by hydrogen). As shown in Fig. 2, synthesis of a five-base run-off transcript on a template encoding GGACU shows the 5-mer as the predominant product using GTP as substrate. In contrast, replacement of GTP by 7-deaza-GTP reduces synthesis of the

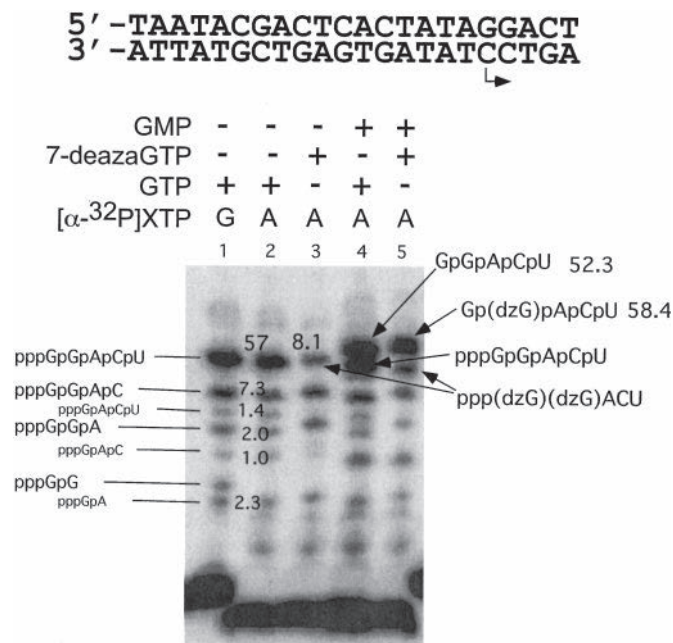


FIG. 2. Run-off synthesis on a template encoding GGACU. GTP, dzGTP, and/or GMP were present at 400 μ M as indicated. All reactions contained 400 μ M each of ATP, CTP, and UTP and 0.2 μ M each of T7 RNA polymerase and promoter DNA. Numbers within the gel represent the amount (μ M) of the transcript formed in a 15-min reaction at 37 °C. Assignment of the bands was carried out by comparing mobilities and, in parallel experiments, the ratios of incorporation of [α -³²P]GTP and [α -³²P]ATP for each band as demonstrated more fully in Fig. 3.

run-off RNA product on this template by more than 7-fold (Fig. 2, compare lanes 2 and 3).

The reduction in transcription in the presence of 7-deaza-GTP could arise from a deficiency in the incorporation of this analog at either the initiating or elongating positions during initiation (or on other templates during subsequent elongation). To distinguish between these possibilities, GMP was added to the reaction. It has previously been shown that GMP can substitute for GTP at position +1, but of course, since it lacks the triphosphate, it cannot substitute at any other positions in the RNA (6). Lane 5 in Fig. 2 shows that the inclusion of GMP in the mixture completely restores normal transcription (note that due to the incorporation of a 5'-terminal monophosphate, run-off transcripts initiating with GMP migrate more slowly on the gel than normal products having a 5'-terminal triphosphate group). This result demonstrates that 7-deaza-GTP incorporates at position +2 as efficiently as GTP but is much less able to incorporate at position +1. These results suggest a role for the N-7 nitrogen of guanine in positioning and/or binding of the initiating NTP. In support of this conclusion, 7-methyl-GTP will also not substitute for GTP in transcription initiation.¹

The N-7 nitrogen of guanine does not appear to be important in elongation synthesis, consistent with previous results that showed that a more dramatic (cyanoborane) substitution at this position is tolerated in dGTP as a substrate for various DNA polymerases (21) and that 7-deaza-guanosine triphosphate can be used during preparative-scale enzymatic synthesis of RNAs more than 30 nucleotides long (22).

The Guanine O-6 Is Also Important for Initiation—Conventional wisdom has been that T7 RNA polymerase prefers to initiate with G but can initiate well with A as the first base in the nascent transcript. This would suggest that the purine O-6 is not important. This view might be supported by the presence

¹ E. Esposito and C. Martin, unpublished.

1: AGGGA 5' -TAATACGACTCACTATA
3' -ATTATGCTGAGTGATATCCCTT

2: GGGAA 5' -TAATACGACTCACTATA
3' -ATTATGCTGAGTGATATCCCTT

DNA template	AGGGA				GGGAA			
GTP	+	+	+	+	+	+	+	+
ATP	+	+	+	-	+	+	+	-
Label, ^{32}P -XTP	αG	αA	γA	αG	αG	αA	γA	αG
	1	2	3	4	5	6	7	8

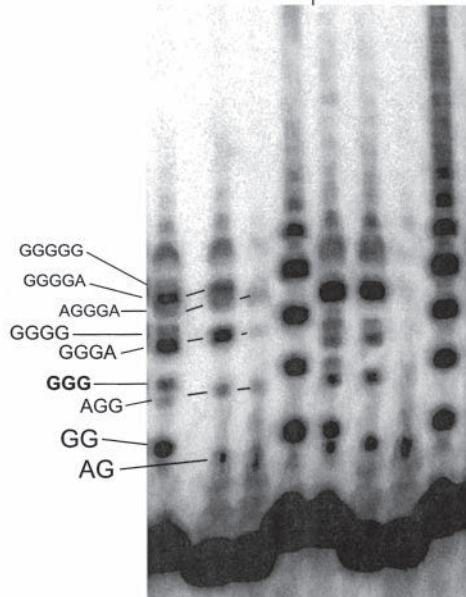


FIG. 3. Transcription from the template encoding the run-off transcript AGGGA compared with that from the control template encoding GGGAA. Reactions contained 400 μM GTP and 400 μM ATP as indicated and 0.2 μM each of T7 RNA polymerase and promoter DNA. Each reaction was run for 10 min at 37 $^{\circ}\text{C}$ as described under "Materials and Methods."

of two promoters in the T7 RNA polymerase genome that encode A at position +1 (9). However, it has recently been shown that on promoters that encode an initial sequence AG . . . , initiation begins primarily with G at position +2 (12, 19). To quantitatively probe this assertion, we have prepared an otherwise consensus promoter construct that encodes the run-off transcript AGGGA. As shown in Fig. 3, lane 1, instead of the clearly defined five-base run-off product dominating the longer products (as in lane 5), a variety of RNA products are synthesized on this template. Comparison of lanes 1–3 demonstrates that despite the encoding of A at position +1, only a minor fraction of products initiate with ATP. The molar distribution of RNA products formed in the presence of GTP and ATP is given in Table I and shows clearly that only about 13% of all RNA products initiate with A; the remainder misinitiate (at position +2) with G. This can be seen not only by comparing the amounts of run-off products but also by comparison of the amounts of the various shorter abortive products.

Interestingly, as shown in lane 4 of Fig. 3 and compared quantitatively in Fig. 4, in the presence of GTP as the sole substrate, T7 RNA polymerase is capable of slippage transcription (23) on the template that encodes AGGGA. Closer examination of the results reveals significant differences between slippage transcription on this alternate template and on the

TABLE I

Identity and amounts of RNAs produced from the template encoding AGGGA (template 1, from data in lanes 1–3 of Fig. 3)

The ratios of incorporated nucleotides aid in the correct assignment of each band to a specific RNA product. The molar concentration of RNA was calculated from the radioactivity and the number of labels expected in each RNA and represents an average of this calculation for parallel experiments with different radioactive labels.

RNA product	$[\alpha\text{-}^{32}\text{P}]\text{GTP}/[\alpha\text{-}^{32}\text{P}]\text{ATP}$	$[\alpha\text{-}^{32}\text{P}]\text{ATP}/[\gamma\text{-}^{32}\text{P}]\text{ATP}$	[RNA] μM
AG	1.1	0.8	1.3
GG			9.5
AGG	1.8	1.0	0.6
GGG			1.1
GGGA (some AGGG)	2.8	14	4.0
GGGG			1.1
AGGGA	1.3	2.0	0.7
GGGGA	3.6		1.9
GGGGG			0.9

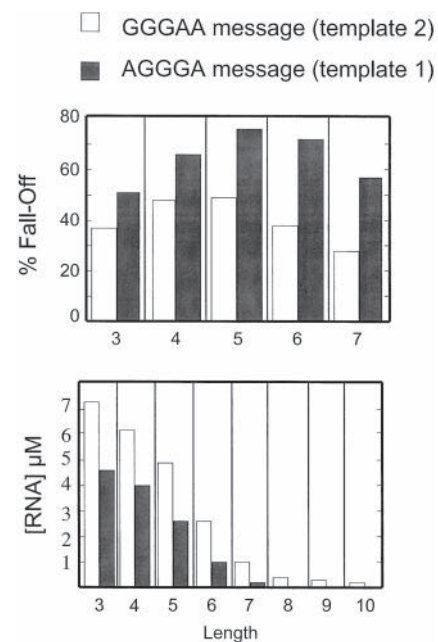


FIG. 4. Comparison of G slippage transcription between two promoters from the data in lanes 4 and 8 of Fig. 3. Percent fall-off is calculated as described under "Materials and Methods."

regular template encoding GGGAA (Fig. 3, lane 8). In particular, in slippage from the AGGGA promoter, there is a consistently higher fraction of fall-off at each step in the slippage. The increased fall-off at each step seen for the AGGGA-encoding template could arise either 1) from a decrease in the rate of forward synthesis at each step or 2) from an increase in the rate of complex dissociation.

Examination of the data in lane 4 of Fig. 3 reveals that there is less RNA produced overall on template 1 (compare with template 2, lane 8 in Fig. 3). This is consistent with the first proposal that misinitiated G slippage leads to a significant reduction in the rate at which the growing RNA chain carries out forward slippage synthesis (in contrast, increased dissociation would lead to higher turnover and larger amounts of RNA).

Estimation of the Binding Strength of 7-Deaza-GTP—To provide better insight into the effect that 7-deaza-GTP has on the kinetics of initiation, we attempted to measure the K_m for 7-deaza-GTP at the initiating (+1) position by using a DNA promoter construct encoding the run-off 4-mer GACU (6). Analysis of the results presented in Fig. 5 turned out to be less than

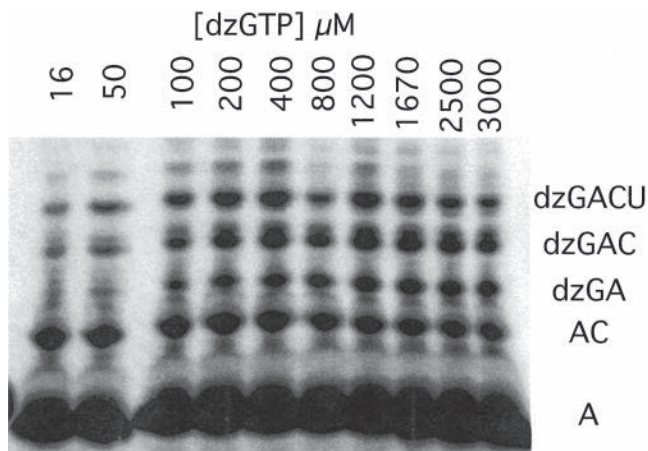


FIG. 5. 7-Deaza-GTP concentration dependence of transcription from a promoter construct encoding GACU. Reactions contained 400 μM each of ATP, CTP, and GTP (with [$\alpha\text{-}^{32}\text{P}$]ATP as label) and were run for 10 min at 37 $^{\circ}\text{C}$.

straightforward. As expected, at very low concentrations of 7-deaza-GTP, misinitiation with ATP (at position +2) predominates, but substantial misinitiation occurs even at high concentrations of 7-deaza-GTP. In addition, extension of the misinitiated products is less efficient as predicted above. As shown in Fig. 5 (and in Fig. 6B), at higher concentrations of 7-deaza-GTP, RNA synthesis (both short and full length) actually decreases rather than leveling off as might be expected. This makes the use of a simple binding equation impossible.

In this context, correct initiation with 7-deaza-GTP competes directly with misinitiation at position +2 (which produces pppApC). Therefore, if the K_m^{ATP} for misinitiation is known, we can use a competitive inhibition model to assess the inhibition constant for 7-deaza-GTP and thereby estimate the strength of its binding at the +1 position. To determine the K_m^{ATP} for misinitiation on this template (in the absence of GTP or its analog), a reaction was carried out measuring initial rates of RNA synthesis as a function of increasing concentrations of ATP. The reaction mixture was identical to the one described above except that it contained 0.4 mM CTP, 0.4 mM UTP, and 10–4000 μM ATP (but no GTP or its analogs). The resulting data, presented in Fig. 6A, were fit to the simple Michaelis-Menten equation (Equation 2),

$$v = \frac{[\text{ATP}]}{K_m^{\text{ATP}} + [\text{ATP}]} V_{\max} \quad (\text{Eq. 2})$$

yielding a value for K_m^{ATP} of $930 \pm 50 \mu\text{M}$.

This value can next be used to fit the data for production of pppApC in Fig. 5 according to the following equation (24).

$$v = \frac{[\text{ATP}]}{K_m^{\text{ATP}} \left(1 + \frac{[\text{dzGTP}]}{K_i^{\text{dzGTP}}} \right) + [\text{ATP}]} V_{\max} \quad (\text{Eq. 3})$$

The best fit of the data, shown in Fig. 6B, yields a value of $K_i^{\text{dzGTP}} = 750 \pm 70 \mu\text{M}$. Interestingly, this value is not substantially different from the K_m of 600 μM determined for initiation with native GTP on the same promoter construct (6), suggesting that the discrimination at position +1 between native GTP and its analog 7-deaza-GTP (dzGTP)² occurs not through a difference in the strength of the binding of the nucleoside triphosphate but rather via effects on some kinetic step following the binding. Clearly, additional studies are required to determine the exact nature of the mechanistic effect.

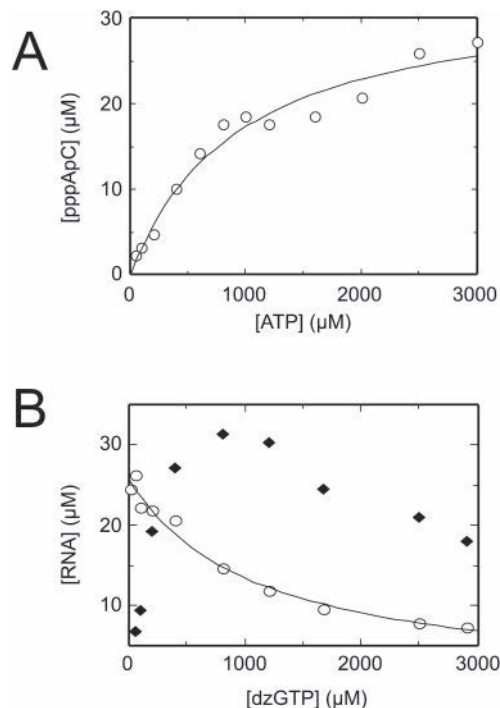


FIG. 6. A, determination of K_m for ATP in the synthesis of the dinucleotide pppApC. B, 7-deaza-GTP concentration dependence of transcription from the promoter construct encoding GACU showing products pppApC (open circles) and all products (filled diamonds). Each reaction was run for 10 min at 37 $^{\circ}\text{C}$. The smooth line represents the best fit to the competitive inhibition model (Equation 2) in which the K_i of $750 \pm 70 \mu\text{M}$ for 7-deaza-GTP represents its ability to competitively inhibit misinitiation at position +2.

DISCUSSION

A critical function of an RNA polymerase is the selection of the position of the transcription start site. From previous studies, it is apparent that a significant part of the energetics stabilizing the initially transcribing complex comes from protein interactions with the upstream duplex region of the promoter (25). Following the separation of the DNA strands (melting), an RNA polymerase must position the first two NTPs in the active site. It seems certain that productive positioning of the first two NTPs in the normally functioning initially transcribing complex occurs a minimal distance from the upstream duplex with this distance determined by the distance along the protein between the tight binding region and the enzyme phosphoryl transfer site (10). Initiation occurs primarily at the 5th and 6th residues downstream of the duplex binding region (19) (and with limited efficiency at the 4th position (13)³).

The enzyme imposes a relatively strict minimal distance, but not a sharp maximal distance, as very large non-nucleosidic linkages still allow initiation at now very “distant” templating bases (26). This accounts for the occurrence of downstream misinitiation when the +1 templating base is a poor initiator either because it does not encode G or because it does not have the optimal environment (e.g. lacking a base 3' of itself).⁴

Hoogsteen Functional Groups Are Important in Positioning the Initiating Nucleotide—Within the part of the DNA where initiation could occur, the choice of initiation seems to be dependent on the preference of T7 RNA polymerase for a particular initiating nucleotide. It is apparent that this preference is not dominated simply by the strength of the Watson-Crick interactions with the complementary base since mutation of

² The abbreviation used is: dz, 7-deaza.

³ I. Kuzmine and C. Martin, unpublished.

⁴ I. Kuzmine, E. Esposito, and C. Martin, unpublished.
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the promoter to encode a C at this position leads to a substantial reduction in activity (19). Indeed, only by using high concentrations of CMP and keeping concentrations of all other nucleotides well below the K_m value for the initiating nucleotide has it been possible to produce substantial amounts of transcripts initiating with C (27). A recent study has implicated the 2-amino group of guanine in such stabilization (12). As the current results show, GTP is preferred over ATP and dzGTP. The latter result implicates the involvement of a major groove contact.

Replacement of nitrogen by CH at position 7 is not expected to substantially reduce the strength of the Watson-Crick base pairing, and although 7-deaza-guanine has been reported to have a lower dipole moment than guanine (3.0 *versus* 8.0) it is oriented in the same direction and seems unlikely to change the geometry of the stacking of the initiating nucleotides (28). The most likely explanation for the preference of the system for initiating with G, and to a lesser extent with A, is the existence of protein contacts with the Hoogsteen side of the initiating nucleotide. Correct positioning of the nucleotide in the active site is likely to involve major groove interactions with both the N-7 and O-6 of the guanine base. ATP and dzGTP each retain only a part of this interaction (N-7 or O-6, respectively) and so are disfavored compared with GTP.

Why Substrate Specificity?—Since the stability of a single base pair is likely insufficient to provide the energetics necessary for correct positioning of substrate, it is easy to imagine how an RNA polymerase would have evolved to provide additional stabilization for binding and positioning of the initiating nucleotide. Note that DNA polymerases, or RNA polymerases during elongation, do not need this type of interaction because the equivalent “initiating” nucleotide is covalently linked to the growing RNA or DNA primer and enjoys additional stabilization through base stacking with its covalently linked 5' neighbor. The problem is unique to *de novo* initiation.

These results suggest major groove interactions with both the N-7 and O-6 groups of guanine. Among the amino acids, the most likely candidate for simultaneous interaction with both the guanine N-7 and O-6 functional groups is arginine (see, for example, Ref. 29). The available crystal structures of T7 RNA polymerase do not allow precise prediction of the Arg in question; however, there are at least two possible candidates: amino acids 425 and 632. These possess a C_α that is less than 13 Å (the approximate length of an extended Arg side chain) from the N-7 of the equivalent G (number 3) in the crystal structure of the ternary complex (Protein Data Bank code 1QLN).

The effect of such stabilization is highlighted by the ability of T7 RNA polymerase to make dinucleotide RNA product on any

long stretch of DNA that has GG or GA encoded in the sequence (3'-CC-5' or 3'-CT-5' in the template), although in the absence of the promoter these products are not extended well (30, 31).⁵

These findings also allow us to speculate on the reason why most of the phage T7 promoters start with the encoding of three G residues at the beginning of the transcribed RNA sequence. In this case, the dinucleotide product formed as the result of misinitiation at position +2 is identical to the product of correct initiation. Since positionally misinitiated products are elongated less efficiently, the dinucleotide is released and can then be reused by priming at the correct position (32).

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⁵ A. Újvári and C. Martin, unpublished.